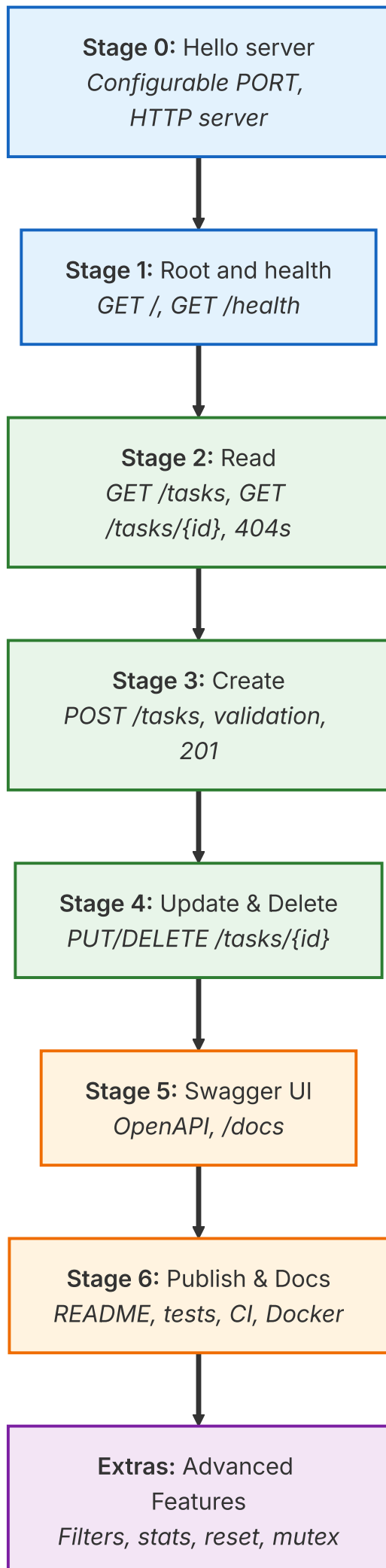


```
type: Assignment
track: Backend AI Engineering
week: 2
phase: Foundations
status: ● Complete
tags:
  - api
  - crud
  - concurrency
  - swagger-ui
  - golang
date: 14/07/2026
name: Build your first CRUD API
code: BE-01
```

Goal

Expand the original two-endpoint Go server into a complete in-memory task API while staying with Go's standard `net/http` package.

Stage mapping



Design decisions

- **Go instead of the two suggested lanes:** the assignment's HTTP and CRUD requirements are language-independent, and the previous project already uses Go.
- **Standard library only:** Go 1.22 supports method-aware route patterns and path wildcards directly in `net/http`.
- **In-memory storage:** the store deliberately resets on process restart, matching the Week 2 learning goal.
- **Pointer fields for updates:** `*bool` distinguishes an omitted `done` field from an explicit `false` value.
- **Strict JSON:** unknown fields, malformed JSON, empty bodies and multiple JSON values are rejected with status 400.
- **Concurrency safety:** a mutex prevents data races when multiple requests access the task list.
- **OpenAPI as a checked-in artifact:** the contract is reviewable at `api/openapi.json` and rendered at `/docs`.

Manual verification checklist

```
go run .
```

In a second terminal:

```
./scripts/demo.sh
```

```

aya@Laptop:/mnt/c/Users/saied/tiny-go-api$ ./scripts/demo.sh
1) Health check
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
Date: Fri, 17 Jul 2026 20:43:57 GMT
Content-Length: 16

{"status":"ok"}

2) List seed tasks
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
X-Limit: 50
X-Offset: 0
X-Total-Count: 3
Date: Fri, 17 Jul 2026 20:43:57 GMT
Content-Length: 159

[{"id":1,"title":"Learn HTTP basics","done":true},{"id":2,"title":"Build a CRUD API","done":false},{"id":3,"title":"Test the API in Swagger UI","done":false}]

3) Create a task
{"id":4,"title":"Buy milk","done":false}

4) Update the task
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
X-Limit: 50
X-Offset: 0
X-Total-Count: 1
Date: Fri, 17 Jul 2026 20:43:57 GMT
Content-Length: 44

{"id":4,"title":"Buy oat milk","done":true}

5) Filter completed tasks
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
X-Limit: 50
X-Offset: 0
X-Total-Count: 1
Date: Fri, 17 Jul 2026 20:43:57 GMT
Content-Length: 46

[{"id":4,"title":"Buy oat milk","done":true}]

6) View statistics
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
X-Limit: 50
X-Offset: 0
X-Total-Count: 1
Date: Fri, 17 Jul 2026 20:43:57 GMT
Content-Length: 30

{"total":4,"done":2,"open":2}

7) Delete the task
HTTP/1.1 204 No Content
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
Date: Fri, 17 Jul 2026 20:43:57 GMT

8) Confirm deletion returns 404
HTTP/1.1 404 Not Found
Content-Type: application/json; charset=utf-8
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
X-Limit: 50
X-Offset: 0
X-Total-Count: 1
Date: Fri, 17 Jul 2026 20:43:57 GMT
Content-Length: 29

{"error":"Task 4 not found"}

```

Then verify:

- GET /tasks/99 returns 404 and JSON.
- POST /tasks with {} returns 400 and JSON.
- DELETE /tasks/{id} returns 204 with an empty body.
- /docs lists and executes the full CRUD cycle.
- After creating tasks, restarting the process restores only the three seed tasks.

```

aya@Laptop:/mnt/c/Users/saied/tiny-go-api$ go test -race ./...
ok      tiny-go-backend 1.033s

```

Mortality experiment

Tasks are stored in a Go slice inside the running process. Restarting the server creates a new store from `seedTasks`, so anything created or updated during the previous run disappears. A database will solve this persistence problem in the next stage of the backend track.